Experiment 7

Design a two conductor transmission line and Investigate the impact of Dielectric layer thickness on characteristic Impedance and Propagation Velocity.

Test Case 1: Design a Two conductor transmission line with single ground plane and estimate the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity. Thickness to be varied between 5 to 15mm,

AIM: To design a Two conductor transmission line with single ground plane and estimate the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity. Thickness to be varied between 5 to 15mm,

OUTPUT-

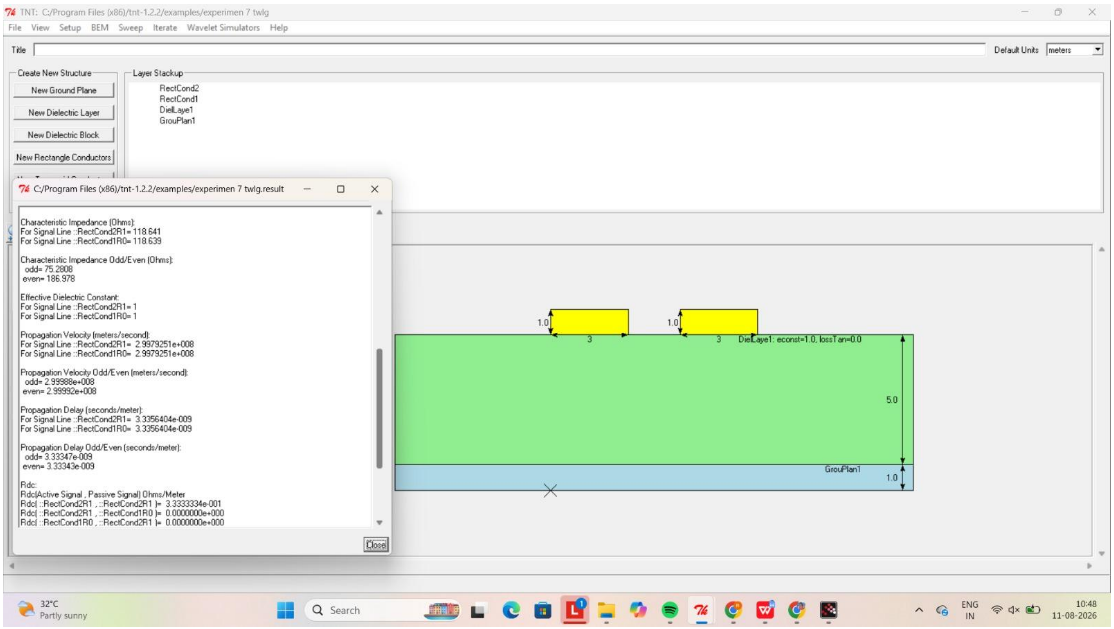

Table-

Thickness of Dielectric layer (mm)	Characteristic Impedance (ohms)	Propagation Velocity (m/s)
5	118.641	2.99e+008
10	142.274	2.99e+008

Test Case 2: Design a Two conductor transmission line with single ground plane and investigate the effect of far end cross talk and near end cross talk by varying the thickness of the dielectric layer .Thickness to be varied between 5 to 15mm

AIM: To design a Two conductor transmission line with single ground plane and estimate the impact of varying the thickness of the dielectric layer on characteristic Impedance and Propagation Velocity. Thickness to be varied between 5 to 15mm,

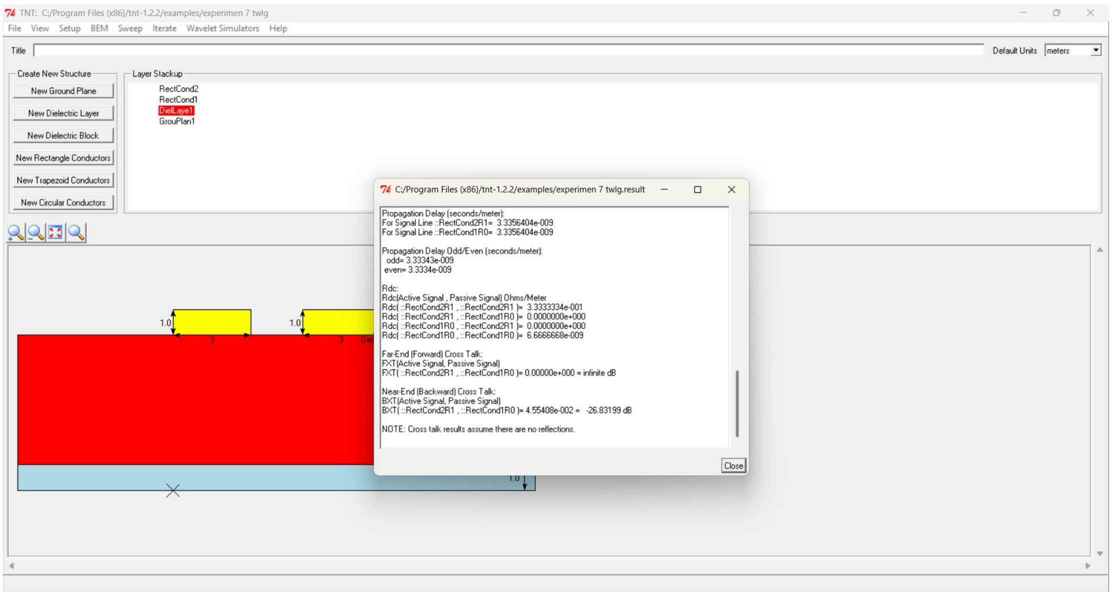

Table-

Thickness of Dielectric layer (mm)	Far end cross talk (dB)	Near end cross talk (dB
5	infinite	-28.850
10	infinite	-26.831